In regression analysis, the main objective is finding the line of best fit, which provides the fitted equation of Y on X.

The line of 'best fit' is the line (straight line equation) which minimizes the error in the estimation of the dependent variable Y, for any specified value of the independent variable X from its range.

The regression equation $E(Y|X=x) = a + bx$ represents a family of straight lines for different values of the coefficients 'a' and 'b'. The problem is to determine the estimates of 'a' and 'b' by minimizing the error in the estimation of Y so that the line is a best fit. This necessitates to find the suitable values of the estimates of 'a' and 'b'.

5.4 METHOD OF LEAST SQUARES

In most of the cases, the data points do not fall on a straight line (not highly correlated), thus leading to a possibility of depicting the relationship between the two variables using several different lines. Selection of each line may lead to a situation where the line will be closer to some points and farther from other points. We cannot decide which line can provide best fit to the data.

Method of least squares can be used to determine the line of best fit in such cases. It determines the line of best fit for given observed data by minimizing the sum of the squares of the vertical deviations from each data point to the line.

5.4.1 Method of Least Squares

To obtain the estimates of the coefficients 'a' and 'b', the least squares method minimizes the sum of squares of residuals. The residual for the ith data point e_i is defined as the difference between the observed value of the response variable, y_i, and the estimate of the response variable, $\hat{y}_i$, and is identified as the error associated with the data. *i.e.*, $e_i = y_i - \hat{y}_i$, $i = 1, 2, \ldots, n$.

The method of least squares helps us to find the values of unknowns 'a' and 'b' in such a way that the following two conditions are satisfied:

- Sum of the residuals is zero. That is $\sum_{i=1}^{n}(y_i - \hat{y}_i) = 0$.
- Sum of the squares of the residuals $E(a,b) = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ is the least.

5.4.2 Fitting of Simple Linear Regression Equation

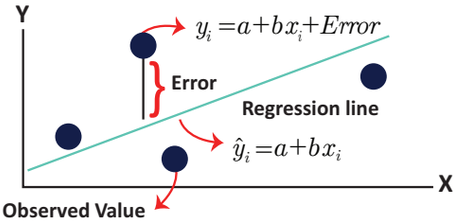
The method of least squares can be applied to determine the estimates of 'a' and 'b' in the simple linear regression equation using the given data (x_1,y_1), (x_2,y_2), ..., (x_n,y_n) by minimizing

$$E(a,b) = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

i.e., $$E(a,b) = \sum_{i=1}^{n}(y_i - a - bx_i)^2 .$$

Here, $\hat{y}_i = a + bx_i$ is the expected (estimated) value of the response variable for given x_i.

Simple Linear Regression Model



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